Thus the algebraic integers are closed under addition and multiplication, and include additive inverses, and of course 0 which is a zero of x. Hence they form a subring of $\mathbb{C}$.

55. Cyclotomic Extensions

1. Let $\zeta = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$. Using the relations

$$\zeta^8 = 1, \quad \zeta + \zeta^7 = \sqrt{2}, \quad \text{and} \quad \zeta^3 + \zeta^5 = -\sqrt{2}$$

we have

$$\begin{aligned}\Phi(x) &= (x-\zeta)(x-\zeta^7)(x-\zeta^3)(x-\zeta^5)\\ &= [x^2 - (\zeta+\zeta^7)x + 1][x^2 - (\zeta^3+\zeta^5)x + 1]\\ &= (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)\\ &= x^4 + 1.\end{aligned}$$

2. The group is $G = \{1, 3, 7, 9, 11, 13, 17, 19\}$ under multiplication modulo 20. We find that $3^2 = 9, 3^3 = 7$, and $3^4 = 1$, so 3 and 7 have order 4, and 9 has order 2. Taking their additive inverses, we see that 17 and 13 have order 4, and 11 has order 2. Of course 19 = -1 has order 2. Because G is abelian, it is isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_2$ by Theorem 11.12.

3. **a.** We have $60 = 2^2 \cdot 3 \cdot 5$, so $\varphi(60) = 2 \cdot 2 \cdot 4 = 16$.

b. Now $1000 = 2^3 \cdot 5^3$, so $\varphi(1000) = 2^2 \cdot 5^2 \cdot 4 = 400$.

c. Factoring, $8100 = 2^2 \cdot 3^4 \cdot 5^2$, so $\varphi(8100) = 2 \cdot 3^3 \cdot 5 \cdot 2 \cdot 4 = 2160$

4. We just use Theorem 55.8 with the Fermat primes 3, 5, and 17, repeatedly multiplying by 2 the numbers $3, 5, 15 = 3 \cdot 5, 17, 51 = 3 \cdot 17$, and $85 = 5 \cdot 17$ until we have generated the first 30 possibilites, which we list in five columns. (The next Fermat prime, 257, is not needed for the first 30 values of n).

3	12	30	60	102
4	15	32	64	120
5	16	34	68	128
6	17	40	80	136
8	20	48	85	160
10	24	51	96	170

5. Now 360 and 180 are divisible by 3^2, so the 360-gon and the 180-gon are not constructible. However, $360/3 = 120 = 8 \cdot 3 \cdot 5$ so the regular 120-gon is constructible, and therefore an angle of 3° is constructible.

6. **a.** We have $[K : \mathbb{Q}] = \varphi(12) = 4$ because the integers ≤ 12 and relatively prime to 12 are 1, 5, 7, and 11.

b. The group $G(K/\mathbb{Q})$ is isomorphic to $\{1, 5, 7, 11\}$ under multiplication modulo 12, and $5^2, 7^2$, and 11^2 are all congruent to 1 modulo 12. Thus for all $\sigma \in G(K/\mathbb{Q})$, we must have $\sigma^2 = \iota$, the identity automorphism. The group is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$ by Theorem 11.12.

7. We have $\Phi_3(x) = \frac{x^3-1}{x-1} = x^2 + x + 1$ for every field of characteristic $\neq 3$, because ζ and ζ^2 are both primitive cube roots of unity.

In $\mathbb{Z}_3[x], x_8 - 1 = (x^4 + 1)(x^2 + 1)(x - 1)(x + 1)$ so the four primitive 8th roots of unity must be zeros of $x^4 + 1$, and $\Phi_8(x) = x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$.

8. In $\mathbb{Z}_3[x]$, we have $x^6 - 1 = (x^2 - 1)^3 = (x - 1)^3(x + 1)^3$, so the polynomial already splits in $\mathbb{Z}_3$, and the splitting field has 3 elements.

9. T T F T T F T T F T

10. The nth roots of unity form a cyclic group of order n under multiplication, for if ζ is a primitive nth root of unity, then

$$\zeta^j \zeta^k = \zeta^{j+k} = \zeta^{j+k \pmod n}.$$

Each element ζ^j of this group generates a subgroup of some order d dividing n, and is thus a primitive dth root of unity. Also, if d divides n, then $\zeta^{n/d}$ is a primitive dth root of unity. Thus the collection of all primitive dth roots of unity for d dividing n contains all the nth roots of unity. Because the primitive dth roots of unity are the zeros of $\Phi_d(x)$ in a field of characteristic not dividing d, we see that $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

11. We have $\Phi_1(x) = x-1, \Phi_2(x) = x+1$, and $\Phi_3(x) = \frac{x^3+1}{x-1} = x^2+x+1$. Also, $x^4-1 = (x^2+1)(x-1)(x+1)$ so $\Phi_4(x) = x^2 + 1$, and $\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$ (see Corollary 23.17). A primitive 6th root of unity is a primitive cube root of -1, and hence a zero of $x^3 + 1 = (x + 1)(x^2 - x + 1)$. Because there are two primitive 6th roots of unity, we see that $\Phi_6(x) = x^2 - x + 1$.

12. By Exercises 10 and 11,

$$\begin{aligned} x^{12} - 1 &= \Phi_1(x)\Phi_2(x)\Phi_3(x)\Phi_4(x)\Phi_6(x)\Phi_{12}(x) \\ &= (x - 1)(x + 1)(x^2 + x + 1)(x^2 + 1)(x^2 - x + 1)\Phi_{12}(x) \\ &= (x^4 - 1)(x^4 + x^2 + 1)\Phi_{12}(x) \\ &= (x^8 + x^6 - x^2 - 1)\Phi_{12}(x). \end{aligned}$$

Polynomial long division yields $\frac{x^{12}-1}{x^8+x^6-x^2-1} = x^4 - x^2 + 1$, so $\Phi_{12}(x) = x^4 - x^2 + 1$.

13. Let ζ be a primitive nth root of unity where n is odd. The positive powers of ζ which equal 1 are then $n, 2n, 3n, \cdots$. Because n is odd, $(-\zeta)^n = -1$. Consequently $(-\zeta)^{2n} = 1$, so the multiplicative order r of $-\zeta$ is either $2n$ or $< n$. If $r < n$, then $1 = [(-\zeta)^r]^2 = [(-\zeta)^2]^r = (\zeta^2)^r = \zeta^{2r}$, and we would then have to have $n = 2r$, contradicting the fact that n is odd. Therefore $r = 2n$, and $-\zeta$ is a primitive $2n$th root of unity.

We have shown that if ζ is a primitive nth root of unity for n odd, then $-\zeta$ is a primitive $2n$th root of unity. Now formula (1) in the text shows that for n odd, $\varphi(n) = \varphi(2n)$, so if $\zeta_1, \zeta_2, \cdots, \zeta_{\varphi(n)}$ are all the primitive nth roots of unity for n odd, then their negatives account for all the primitive $2n$th roots of unity. Now ζ is a zero of $\Phi_n(x)$ if and only $-\zeta$ is zero of $\Phi_n(-x)$, which, by Definition 55.2, shows that $\Phi_{2n}(x)$ must be either $\Phi_n(-x)$ or $-\Phi_n(-x)$, depending on whether the degree $\varphi(n)$ of $\Phi_n(x)$ is even or odd. But formula (1) in the text shows that if n is odd, then $\varphi(n)$ is even. Thus we have $\Phi_{2n}(x) = \Phi_n(-x)$.

14. Let ζ be a primitive mnth root of unity. Then ζ^m is a primitive nth root of unity and ζ^n is a primitive mth root of unity, so the splitting field of $x^{mn} - 1$ contains the splitting field of $(x^m - 1)(x^n - 1)$.